

“**Indian OEMs are developing portable devices like digital thermometers, blood pressure monitors, insulin pumps, heart rate monitors and digital hearing aids, in addition to hospital equipment like CT and**

MRI scanners—Rakesh Joshi, business development manager (medical business), Texas Instruments India



are now helpful for pre-natal foetal examination. Improved resolution of image is another advancement.

Patient monitoring and management systems. These help measure parameters that are integral to the physiological functioning of the body; for example, EEGs and ECGs. They are also used to monitor and maintain saturated pressure of oxygen and other such parameters; for example, respiratory chambers and the like recently used in the treatment of the H1N1 virus.

Segments like ECG and neuro-physiological monitoring systems have developed rapidly and continue to advance due to higher adoption rates.

Wireless lower-acuity monitors that are the size of a cell phone can be used with wireless local-area network (WLAN) to transmit the vital signs of emergency room patients who might be ambulatory or moved to other areas of a hospital.

Higher-acuity monitors too can be wireless but due to the size, memory and processing power limitations of wireless technology, these monitors typically transmit fewer measurements than standard, non-wireless monitors.

Devices and implants. The total demand for cardiac implants will continue to rise largely due to the widening use of cardiac resynchronisation therapy (CRT) devices in the treatment of congestive heart failure (CHF). CRT devices are more effective than alternative drug therapies in restoring the blood pumping capabilities of CHF patients along with reducing their risk of disease complications. Ultra-low-

power electronics finds applications in this area. Use of RF technology to control the electrical stimulation for neurological stimulators is one trend.

In defibrillators, use of devices with biphasic technology and low-energy defibrillation is the emerging trend. Multiple-direction current in biphasic defibrillators lowers the threshold for successful defibrillation, saving more lives. Biphasic defibrillators adjust the waveform as needed based on the patient to deliver the best possible waveform at any given time.

Telemedicine. Telemedicine and e-healthcare have immense potential for growth in a country like India where a major portion of population resides in rural India with very little or no access to quality healthcare advice and treatment. Additionally, with an increase in lifestyle related diseases like diabetes and cardiac diseases, it becomes obvious that we need easily accessible technology and a common platform that connects billions of patients spread across the country to medical experts in time of need.

Embedded technologies and software. The function for embedded systems in medical technology is never-ending. For example, new embedded systems are being developed where pills are taken that are ready with

smart processors to repair organs or process information about cell formation or irregularities in cell operation. Now, this is a small industry, but it is possible that embedded systems can do any medical processes in future, like surgeries or analysis of X-rays.

Chips bringing about revolution

Vivek Tyagi, country sales manager, Freescale Semiconductor India, believes that semiconductor technology will play a critical role in the development of new technologies that assist in patient monitoring, diagnostics, therapy and imaging.

The need today is to have products that offer ultra-low-power consumption, driven by the need for extended battery life, and are highly efficient and high-precision with a fast response time. All this has been possible with ICs. ICs have a significant impact on the end usage. These can help with miniaturisation and lower power consumption. New-generation processors can improve the user experience.

In portable medical equipment, the core components must meet tight requirements like low operating voltage, low operating current, low standby current, small packaging options, on/off power management, low noise, high-efficiency step-up and multiple battery topologies. ICs are being designed to meet these standards as required.

Electronic design automation (EDA) software is used by semiconductor companies to design chips for medical electronics applications like scanners, portable medical meters and medical imaging solutions. These chips are required to have very high reliability and precision. For mobile devices,

“**Devices are becoming affordable and portable aided by the fact that chips are being packed with more functionalities**—Sridhar Perepa, vice president-R&D services, MindTree

